

State Tracking in LLMs: Models Recall the First Value but Lose the Current State

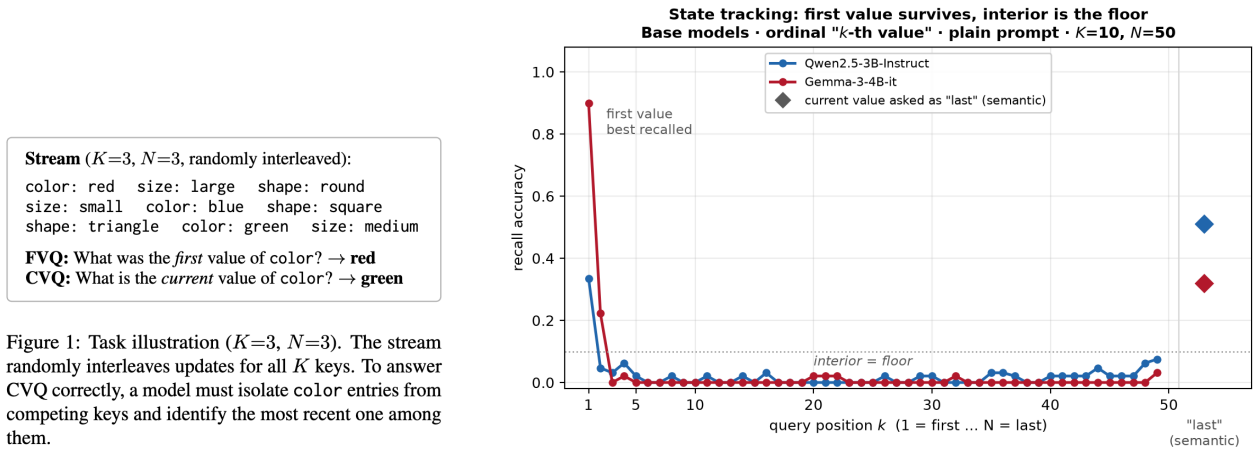
Kanak Raj · kanak8278@gmail.com

Mechanistic extension of the ACL paper “Transformers Remember First, Forget Last: Dual-Process Interference in LLMs.”

The Question: Can an LLM Track the State of a Variable?

State tracking: a key’s value is set, then updated repeatedly; we query its value at a chosen update position k . A stream interleaves updates to K keys, each updated N times. The first ($k=1$) and current ($k=N$) values are the endpoints; the real test is the interior.

The finding: models reliably return the first value but not the interior or current state — tracking works at the edges and collapses to the floor in between. Sweeping k (right): first-value recall is best; by $k=3-5$ accuracy is at chance and stays there. The current value is recoverable only via the semantic anchor “the last value” — asked ordinarily (“the 50th value”) it collapses too. The model holds an anchor at each end, not per-position state.



1. Even the Two Endpoints Diverge — Universally, and Not from Context Length

Restricting to just the two endpoints (first vs. current) shows the asymmetry in every model tested, proprietary and open-weight. And it is not a context-length problem: every failure below occurs while the prompt occupies under 3.3% of the model’s advertised window.

Model	FVQ	CVQ	Gap	% Ctx
<i>Proprietary models, $K=20, N=50$</i>				
GPT-4.1	1.00	0.71	+0.29	0.7%
GPT-4.1-mini	1.00	0.74	+0.26	0.7%
Claude-4.5-Haiku	1.00	0.53	+0.47	3.3%
Claude-4.5-Sonnet	1.00	0.74	+0.26	0.7%
Gemini-2.5-Flash	1.00	0.54	+0.46	0.6%
Gemini-2.5-Pro	0.90	0.77	+0.13	0.7%
<i>Open-weight models, $K=5, N=30$</i>				
Qwen2.5-3B-Instruct	0.61	0.55	+0.06	3.1%
Qwen3.5-2B	0.62	0.48	+0.14	0.4%
Qwen3.5-4B	0.86	0.64	+0.22	0.4%
Qwen3.5-9B	0.86	0.79	+0.07	0.4%
Gemma-3-1b-it	0.55	0.02	+0.53	3.1%
Gemma-3-4b-it	0.97	0.68	+0.29	0.8%

Table 1: FVQ and CVQ at the smallest cell per block where every model has gap $> +0.05$ ($K=20, N=50$ proprietary; $K=5, N=30$ open-weight). % Ctx is the prompt’s share of the model’s advertised context window. No model uses more than 3.3% of its window at the cell it fails on.

2. Degrades with Interference, Not Context Length

The current-value accuracy decays as either the number of keys K or updates per key N grows, while first-value recall stays pinned near 1.0. More competing state — not a longer prompt — is what breaks tracking.

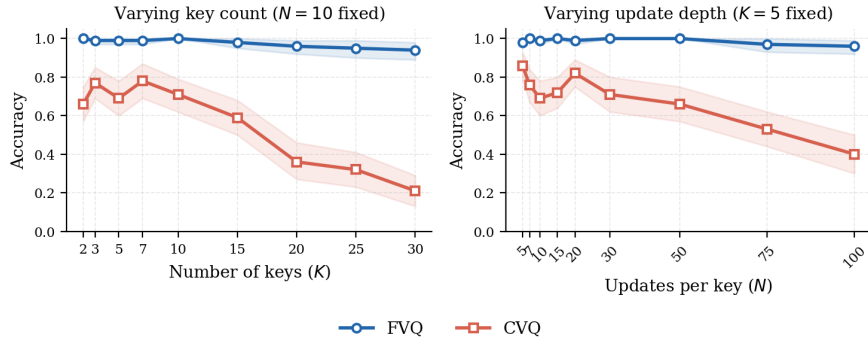


Figure 2: Gemma-3-4b. First value (blue) near ceiling across both axes; current value (red) falls to ~ 0.21 at $K=30$ and ~ 0.40 at $N=100$.

3. Present at Every Training Stage — No Stage Causes or Cures It

Tracking the endpoint gap across full training trajectories (SmoLLM2-1.7B, SmoLLM3-3B) shows it is present at the earliest checkpoints and persists unchanged through every stage — pretraining, SFT, mid-training, and alignment. No single stage introduces or removes it; it is not an instruction-tuning artifact.

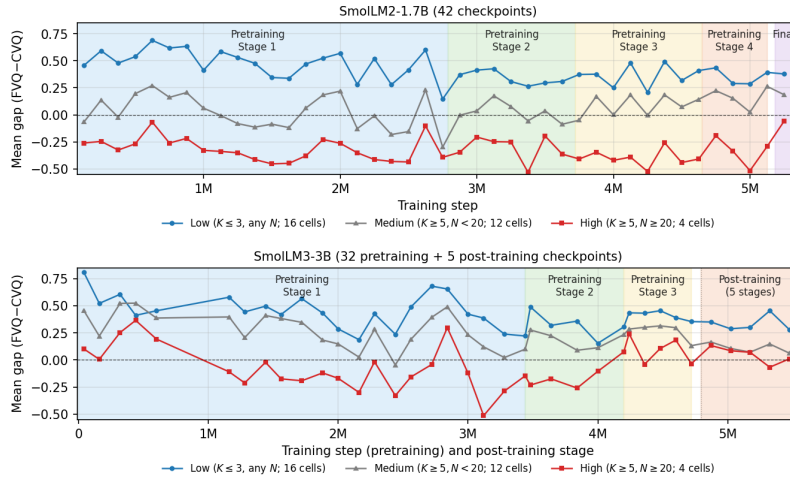


Figure 3: Mean first–current gap across 42 (SmoLLM2-1.7B) and 32+5 (SmoLLM3-3B) checkpoints. Positive from the first checkpoints through all pretraining and post-training stages.

4. Mechanism: the Correct State Is Computed, Then Thrown Away

Logit lens shows where tracking breaks. On current-value failures, the probability of the correct value rises to a peak at $\sim 85\%$ network depth — the model finds it — then is suppressed in the final layers, outcompeted by the penultimate value. For the first value, probability rises monotonically to near 1.0 with no competition. The failure is a late-layer representational collapse, consistent across architectures.

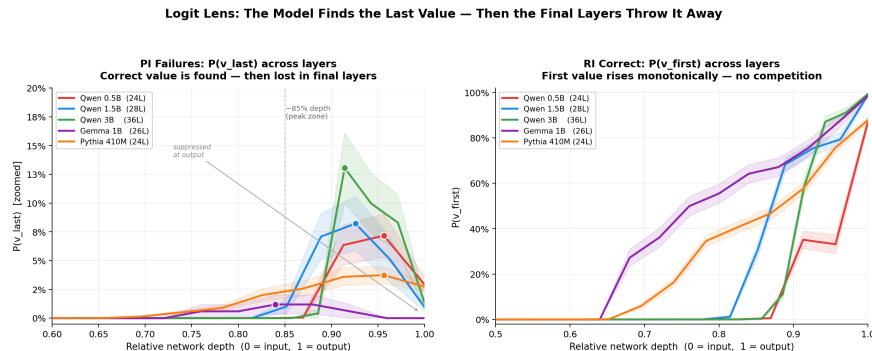


Figure 4: Left — current-value failures: $P(v_{\text{last}})$ peaks at 81–92% depth across 5 models, then collapses at the output. Right — first-value correct: $P(v_{\text{first}})$ rises monotonically, no suppression.

5. It Survives in Natural Language, on a Frontier Model

The effect is not a synthetic-format artifact. On naturalistic museum-visit narratives, Claude Haiku 4.5’s recall decays from a primacy peak straight to the floor with no recency bump; recency returns only through an explicit “last” anchor — the same interior-floor signature.

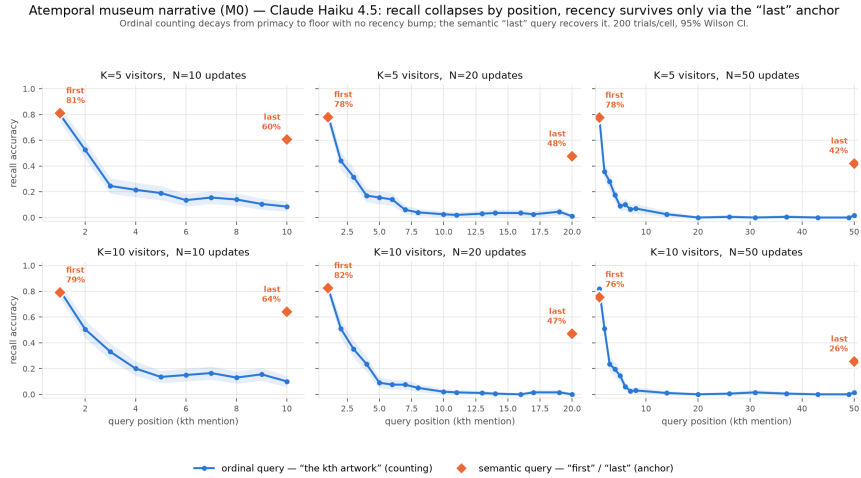


Figure 5: Claude Haiku 4.5 on naturalistic narratives (200 trials/cell). Ordinal recall (blue) decays from ~80% at “first” to floor; only the semantic “last” anchor (orange) partially recovers the current state.

6. Can We Fix Interior Tracking? Format Helps; Fine-tuning Does Not Generalize

The state is present — a block layout (grouping each key’s updates) reads it back almost perfectly, lifting interior accuracy from the floor (~0.06) to ~0.93–0.97. LoRA fine-tuning on the plain format also lifts the interior, but only partway, and it decays out-of-distribution: it learns anchors and near-training cells rather than the general “count to the current value” operation. Concretely, LoRA-adapted Qwen2.5-3B and Gemma-3-4B answer first/last-value queries correctly even far outside their training range, yet still fail on mid-chain (interior) queries at those same large cells (here “first”/“last” are the semantic queries — “the first/last value” — not a numeric k -th position).

Model (interior IVQ)	Base, plain	Base, block	LoRA, in-dist	LoRA, far-OOD
Qwen2.5-3B-Instruct	0.06	0.93	0.55	0.25
Gemma-3-4b-it	0.06	0.97	0.43	0.24

Table 1: Mean interior (ordinal) accuracy. Plain-format base is at the floor; a block layout nearly solves it (the state is recoverable); LoRA lifts it in-distribution but only partially, and regresses on far-OD $K \times N$ cells.



Figure 6: LoRA generalization: near-ceiling in-distribution and near extrapolation (A), but far-OOD interior cells collapse while the first/current endpoints hold (B).

7. It Is Intrinsic to the Objective — a From-Scratch Model Reproduces It

A GPT-2-small trained from scratch on the task reproduces the same structure. Within each panel accuracy is U-shaped — the first and last positions are easiest, the middle hardest — and adding keys (K) pushes the whole curve down. At long histories, low-to-mid K stays strong while high K collapses to near-chance across every position at once. Held-out (never-trained) cells (stars) behave like the trained ones, so this is generalization, not memorization. The bias is not inherited from web-scale data — it emerges from the task and architecture.

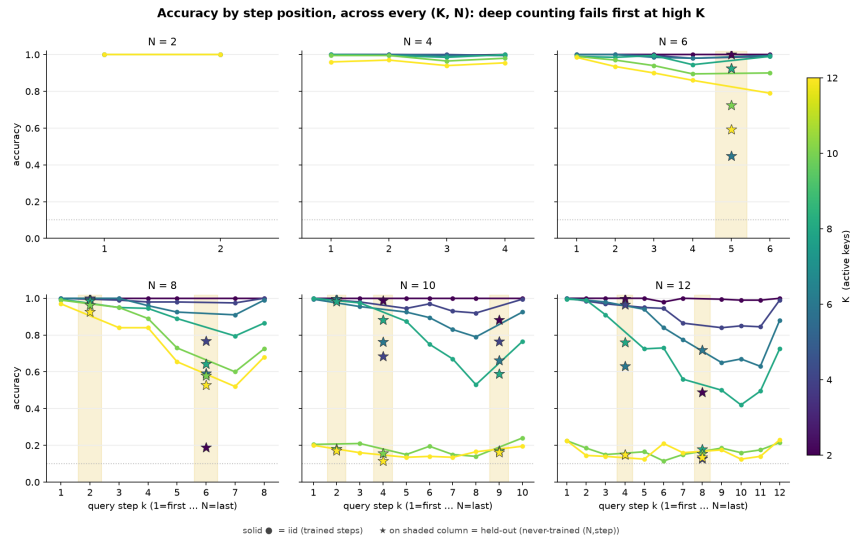


Figure 7: GPT-2-small from scratch: accuracy by query position k for each (K, N), colored by number of keys K . Within a panel the curve is U-shaped (first/last easiest, middle hardest); more keys push it down; at long N , high K collapses across all positions. Stars mark held-out (never-trained) cells.